

Zoom lens

A zoom lens provides a range of focal length (say the 18-55mm range from our previous example) that the photographer can vary in sharp contrast to the “prime” or fixed focal length lens where it can’t be done. The primary advantage of a zoom lens is that it’s easier to achieve a variety of compositions or perspectives (since lens changes aren’t necessary) without changing your physical position which might not be either convenient (say, in a party) or possible (say, in wildlife).



A telephoto fixed aperture lens. Canon 300mm f/2.8

For an ultra-long range like 200mm+ focal length, you typically have two options. One is the variable aperture zoom lens and the other is the fixed aperture (Either at f/2.8 for mid-range telephoto or f/4 for ultra telephoto) lens. While the former ones are much more inexpensive they reduce the aperture at the telezoom end of the focal length hence allowing far less light to come in, which results in slower shutter speeds, which in turn will translate into blurry images without proper stabilisation.

On the other hand, the fixed aperture telezoom lenses (The ones you see with sports or wildlife photographers) are much more expensive (Often at least 10 times), far heavier in weight with huge dimensions as compared to variable aperture zoom lenses. The advantages are obvious in superior optical performance and the ability to open up to f/2.8 even at focal lengths upto 300mm.

Prime lens

A prime lens is a photographic lens whose focal length is fixed, as opposed to a zoom lens, so you can’t zoom in or zoom out to change the composition without physically moving yourself. This use of the term “prime lens” is an Class A exhibit of the term retronym. Earlier, only prime camera lenses were available, and were merely called “lenses” or “objectives”. Later, “auxiliary” lenses were available, which usually fit in front of the front element of the primary or “prime” lens.

One may question why someone would intentionally restrict their options by using a prime lens? Well, it’s a valid concern and the answers are equally important. For the same price, a prime lens offers much supe-